User Manual

Methods Validation App

Version 1.1

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1 Introduction

This User Manual describes the functionalities, workflow, and usage of the web-app version of the Methods Validation App (MVA). The aim is to provide clear and practical instructions for end-users.

The application is freely available at https://mva.databloom.it/.

1.1 Purpose of the Application

The Methods Validation App is designed to simplify and standardize the evaluation of analytical methods by automating the statistical procedures required during method validation. Its purpose is to provide analysts with a reliable, user-friendly environment in which calibration models, accuracy, precision, detection limits, and other key validation parameters can be computed consistently and reproducibly.

MVA supports validation strategies aligned with the main international standards, including the ICH Q2(R2) guidelines. Its workflow and statistical framework make it suitable for applications that must comply with regulatory expectations in pharmaceutical, clinical, forensic, environmental, and other analytical contexts.

By integrating data handling, statistical testing, and graphical inspection tools in a single interface, MVA reduces manual calculation errors, improves reproducibility, and helps laboratories maintain high analytical quality standards.

1.2 Target Audience

This manual is intended for analysts, laboratory technicians, quality assurance personnel, and researchers who perform or supervise the validation of analytical methods. MVA is suitable for users with varying levels of statistical expertise, from beginners who need guided workflows to experienced analysts who require a fast and reliable tool to support validation.

Getting started quickly

If you just want to explore the app, open the *Import data* page and press **Load example dataset**. A ready-made Amphetamine calibration curve is loaded, so you can walk through every module without preparing a file first.

1.3 How To Cite

If you use this software, cite it as: [CITE TO ADD]

2 Architecture

MVA is structured as a browser-based application that separates the user interface from the statistical computation layer. The graphical interface, built with NiceGUI, manages the interaction with the user and orchestrates the exchange of data with the underlying Python engine. Users load datasets, select validation parameters, and inspect outputs through the frontend, while all analytical operations (regression modelling, variance assessment, residual analysis, and evaluation of precision and accuracy) are executed exclusively in the backend.

The backend consists of a collection of Python modules that implement the statistical framework described in the validation protocol. The file **stat_test.py** contains the core computational logic. It receives the normalized dataset prepared by the interface and performs the entire regression-selection routine. The workflow begins with the grouping of calibration levels and the evaluation of homoscedasticity using both the Levene test and an F-test. The outcome of these tests determines whether the backend will construct models using ordinary least squares or weighted least squares. When heteroscedasticity is detected, the backend computes three candidate weighting schemes (no weighting, $1/x$, and $1/x^2$) and selects the optimal one by comparing the variance of normalized residuals across calibration levels.

Once the variance structure has been established, the backend fits both linear and quadratic regression models. The implementation uses the Ordinary Least Squares (OLS) and Weighted Least Squares (WLS) interfaces from *Statsmodels*, generating paired models for the averaged calibration points as well as for the raw data. Model selection relies on a sequence of diagnostic tests: Mandel's test verifies whether the quadratic term provides a statistically significant improvement, and a secondary check compares paired residuals using a t-test and an F-test. Only when both tests confirm the absence of meaningful differences does the backend override Mandel's result in favour of the simpler model. Residuals from the selected model undergo the Shapiro-Wilk test, and a kernel density estimate is optionally computed for visualization purposes.

The backend also implements the Hubaux and Vos approach for the estimation of the detection and quantification limits, cast in the CCα/CCβ formulation of del Río Bocio et al. (J. Chemometrics 17, 2003). Building on the Hubaux and Vos prediction band around the low calibration levels (Alladio et al., MethodsX 7, 2020), the backend solves in closed form for the *decision limit* CCα (the lowest signal distinguishable from the blank at risk α) and the *detection limit* CCβ, i.e. the LOD, read off the lower prediction band. To keep both estimates *conservative*, the intercept and slope of the working line are taken from the **unweighted (OLS) fit**: a variance-optimal weighting would tilt the line and shrink the residuals near $x = 0$, where the limits are read, and would report artificially smaller values. The chosen weighting scheme still enters the error-propagation term of the prediction interval. The implementation supports internal-standard-normalized data and adapts the weighting computations accordingly.

Precision and accuracy calculations follow a leave-one-out back-calculation route. The backend reorganizes the dataset by days and calibration curves, builds the admissible combinations of calibration curves, constructs the corresponding model, and predicts the left-out curve(s). Precision is derived from the coefficient of variation across predictions, and accuracy from the mean back-calculated concentrations relative to the nominal values. Both intra-day and inter-day routines reuse the same modelling functions and statistical tests, ensuring full methodological consistency throughout the validation process.

Data exchange between the interface and the backend occurs in the form of structured Python objects, which the frontend translates into tables or plots. The frontend never performs statistical operations; it merely triggers backend routines and displays their outputs. Each browser session keeps its own dataset in memory, so results are private to the user and can be cleared at any time.

3 User Interface Guide

3.1 About page

The About page (Figure 1) serves as the application's homepage and is the first interface displayed upon launching MVA. It provides a concise yet informative overview of the software's purpose, guiding principles, and core functionalities. The page introduces users to the rationale behind the implemented validation workflow, outlining how MVA structures and automates key analytical steps such as calibration, precision, accuracy, and sensitivity assessments.

In addition to the introductory text, the About page includes a visual timeline summarizing the sequence of validation tasks supported by the application **B**. This helps users quickly understand the overall workflow. A dedicated bibliography section **C** is also provided, featuring the key references that form the scientific and methodological foundation of MVA. On the left, the collapsible drawer menu **A** gives access to every module, while **Clear User History** **D** wipes the current session from memory.

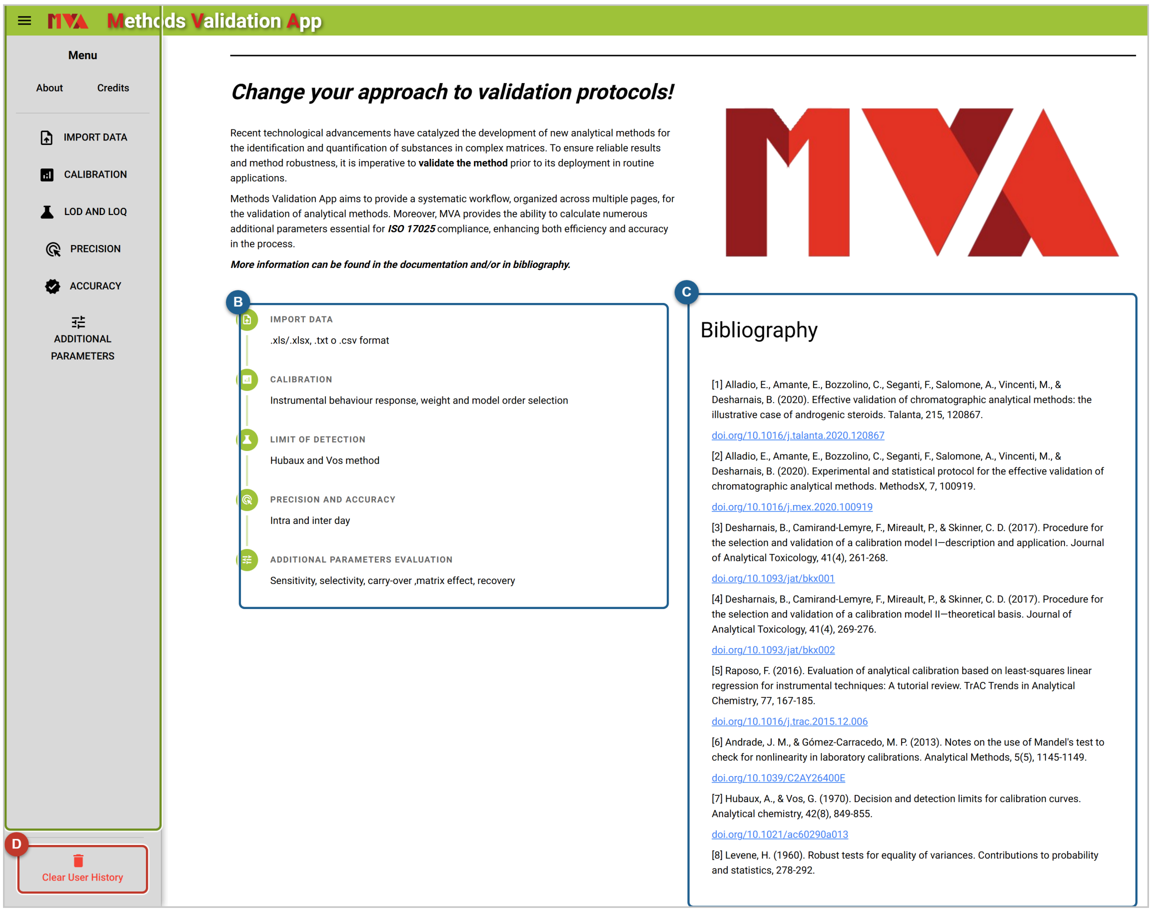

Figure 1. About page of the Methods Validation App.

3.2 Navigation Menu

The closable left-drawer menu (Figure 2) provides access to all sections of the application and allows users to navigate seamlessly between the different MVA modules. At the top, the **About** [1] and **Credits** [2] shortcuts open the two informational pages. Below them, the six analysis modules are listed: *Import data* [3], *Calibration* [4], *LOD and LOQ* [5], *Precision* [6], *Accuracy* [7], and *Additional parameters* [8]. At the bottom, **Clear User History** [9] removes all data stored for the current session.

Each calculation page (Calibration, LOD and LOQ, Precision, Accuracy, and Additional parameters) operates independently, meaning that analyses performed in one module do not influence or modify the results of another. Data uploaded through the *Import data* page serve as the foundational dataset for every calculation module with the exception of *Additional parameters*. This latter section requires its own dedicated input files, as the computations performed there involve parameters and experimental layouts that differ from those used in the main validation workflow.

3.3 Credits page

The Credits page (Figure 3) acknowledges the collaboration between **DataBloom** and the **University of Turin** behind MVA. It lists the individual contributors with their affiliation and contact address, summarizes the licensing terms (MVA is distributed as freeware for academic and research use, with a professional version available on request), and offers a *Report a Bug* button to notify the development team of unexpected behaviour or feature requests.

4 Workflow and Functionalities

4.1 Data Import

The *Import data* page (Figure 4) is the starting point of the analytical workflow and provides all tools required to upload, inspect, and prepare datasets before running any calculation module. The page supports files in `.csv`, `.txt`, `.xls`, and `.xlsx` formats, which are validated on upload to ensure compatibility with the processing routines.

Three entry points are available: the **upload area** [1], where a file can be dropped or selected; the **Info** button [2], which describes the expected data layout and offers a downloadable template; and the **Load example dataset** button [3], which loads a bundled Amphetamine calibration curve (five levels, three curves per day over three days) and pre-fills the whole form so the app can be tried without preparing a file.

Once a file is uploaded, the system automatically parses the content and displays the available column names.

Data layout

Column names must be on the **first row** of the imported dataset. Enter the data day by day, listing all calibrators of curve A first, then curve B, and so on. Each calibration curve is labelled with a letter (A to I), and the same number of curves must be used on every day.

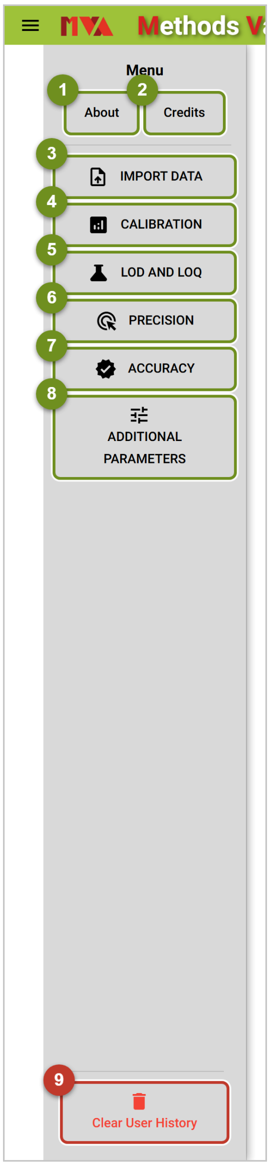

Figure 2. The navigation drawer, with its shortcuts, analysis modules and session control.

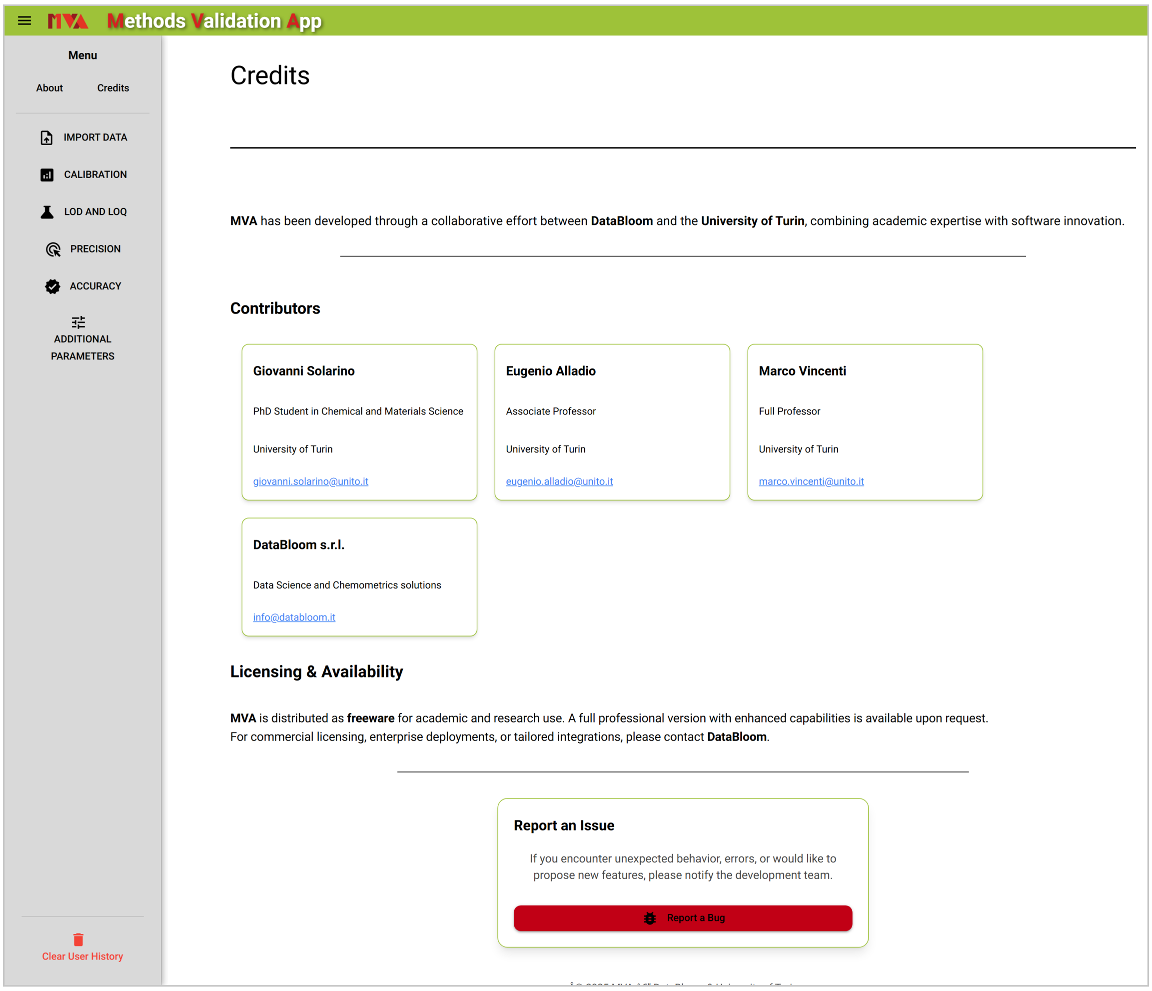

Figure 3. Credits page: contributors, licensing information and the bug-report entry point.

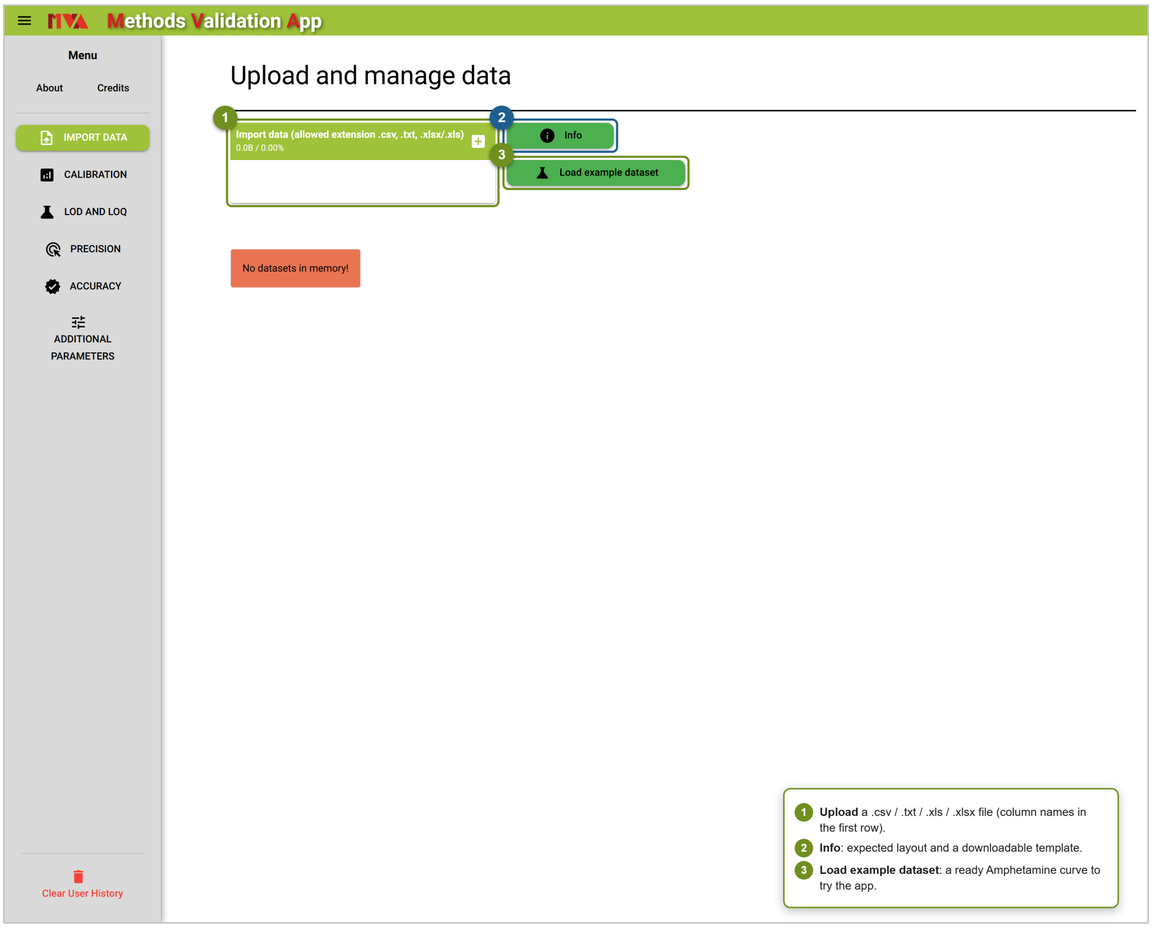

Figure 4. Import data page: upload area, template information and the example-dataset shortcut.

Users are then prompted to map three essential columns and to provide three metadata fields (Figure 5):

- the **concentration** column 1;
- the **analyte signal** column 2;
- the **internal standard (ISTD) signal** column 3;
- the **analyte name** 4, **measurement unit** 5, and **ISTD concentration** 6.

These selections determine how the dataset is normalized and structured before being stored. The app performs an internal preprocessing step that includes signal normalization (when an internal standard concentration is provided), consistency checks, and error handling to prevent misalignment between selected fields. If duplicate selections or incompatible columns are detected, a notification alerts the user and prevents further processing.

No internal standard?

If an internal standard is not available, fill the ISTD signal column with a list of 1s. The analyte signal will then be used directly, without normalization.

When all fields are defined, the **Show data** button 7 becomes available. Pressing it creates a normalized dataframe preview (Figure 6), which is displayed in a table together with a **Clear Data** switch 1 and a **Go to Calibration** button 2. The processed dataset is then saved in memory and made accessible throughout the application. The status panel at the bottom of the page indicates whether a dataset is currently loaded and offers options to *Change analyte* (remap columns on the same file) or *Clear memory* entirely. A template dataset is also available through the **Info** button. This template illustrates the expected data structure. It can be downloaded directly for reference or for preparing new datasets.

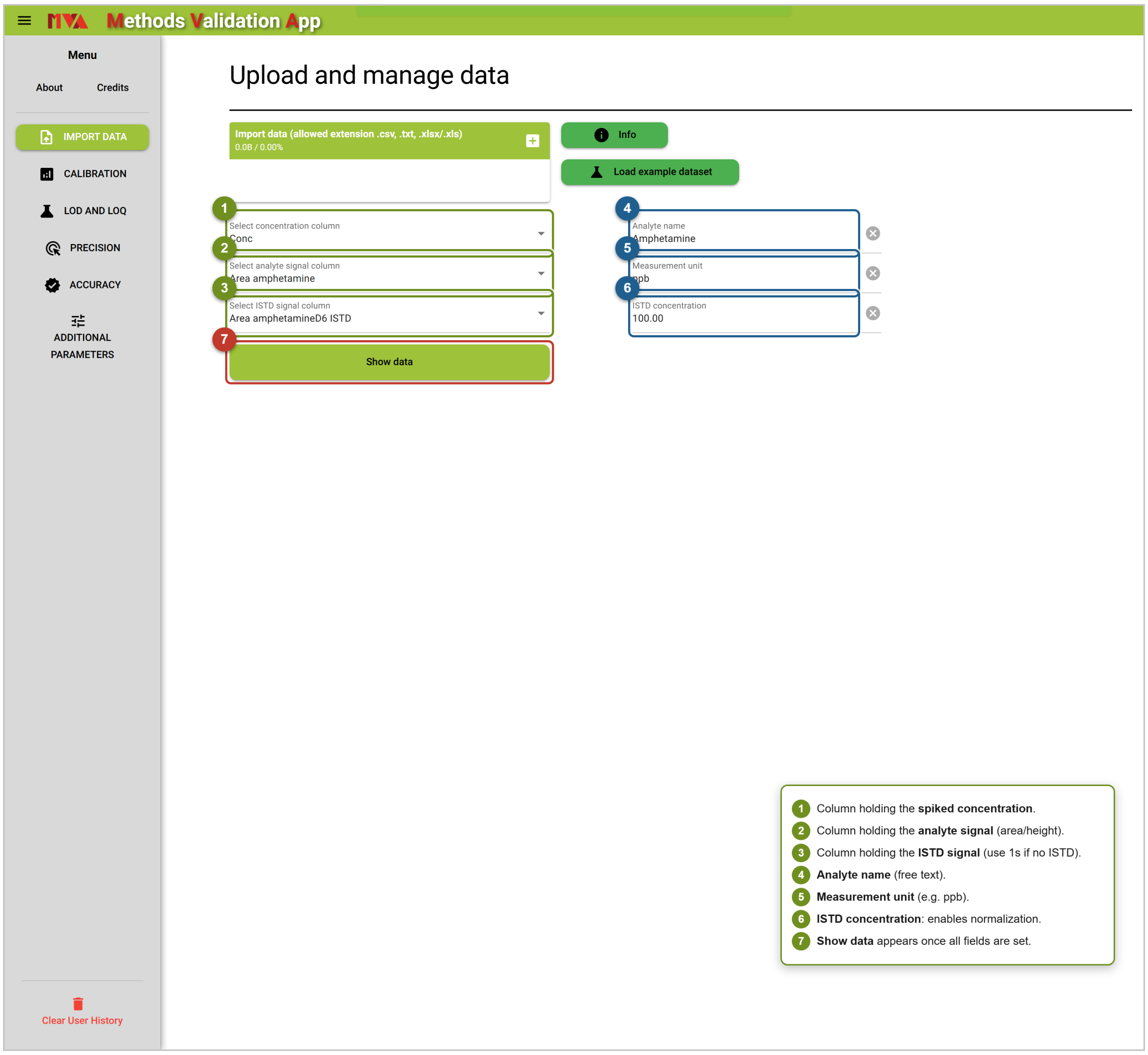

Figure 5. Column mapping and metadata: once every field is set, the *Show data* button appears.

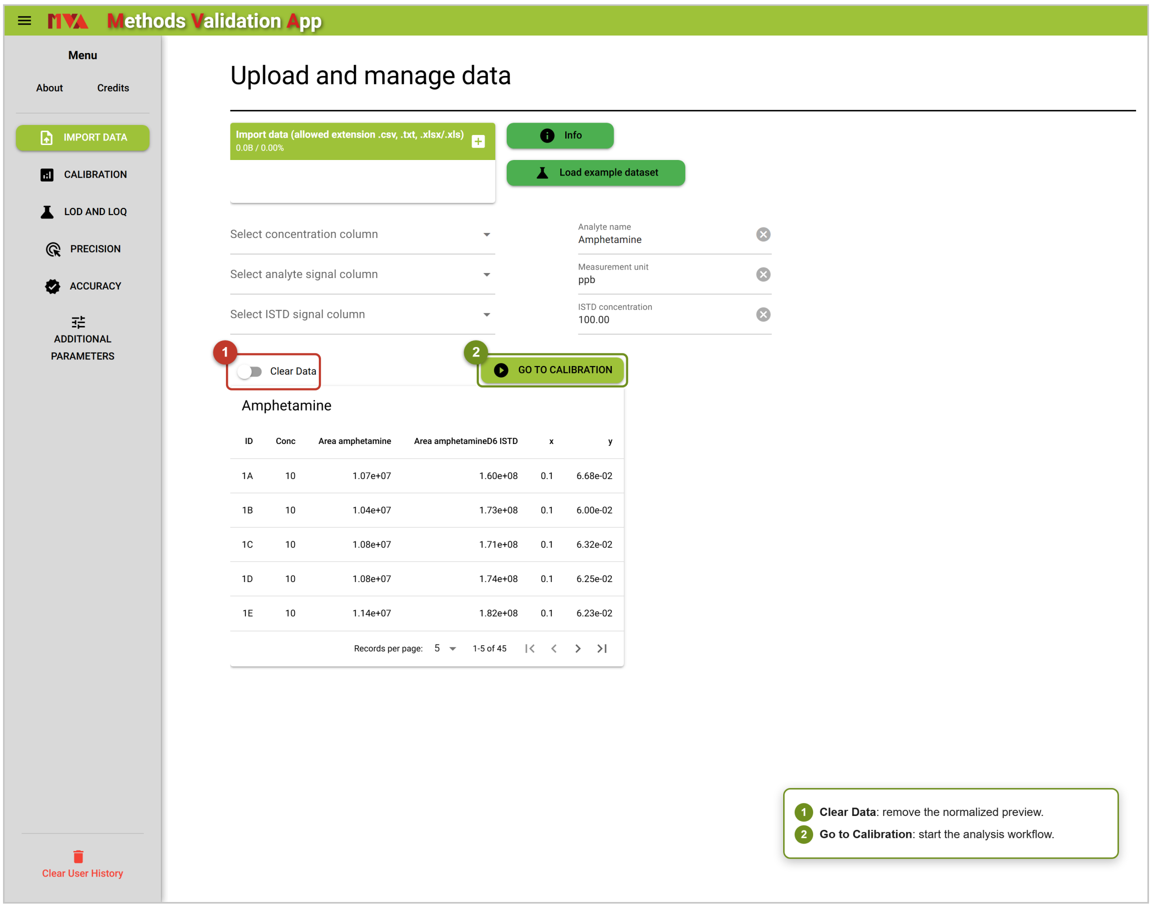

Figure 6. Normalized dataframe preview, with the added columns x (concentration ratio) and y (signal ratio).

4.2 Calibration study

After clicking **Go to Calibration**, the user is redirected to the *Calibration* page, where the calibration study is performed. The imported data are first displayed both in tabular format and as a scatter plot (signal ratio vs. concentration ratio) showing all replicates and their level means (Figure 7). Most figures are interactive: hovering reveals values, and the toolbar allows zooming and downloading.

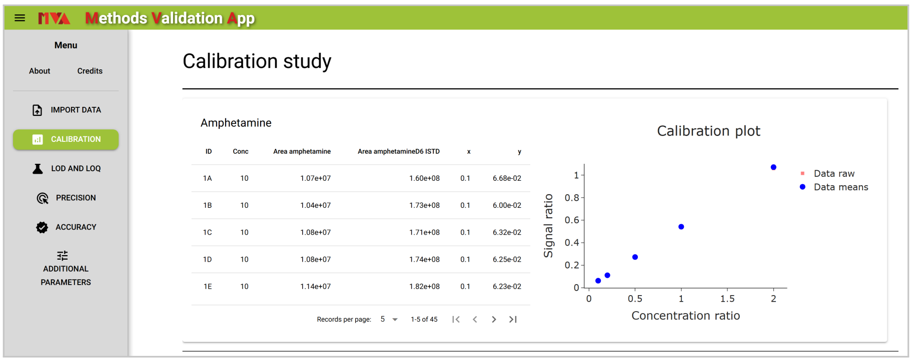

Figure 7. Calibration page: imported data shown as a table and an interactive scatter plot.

Scrolling down, the user finds a set of tab panels dedicated to heteroscedasticity testing (Figure 8). The **Levene test** 1 and the **F-test ULOQ-LLOQ** 2 present the corresponding statistical results and interpretations; for the F-test, a graphical representation of the data dispersion at the Upper and Lower Limits of Quantification (ULOQ, LLOQ) is also provided. The **Weight selection** panel 3 is populated only when heteroscedasticity is detected, and reports the weighting factor ($1/x$, $1/x^2$ or none) that minimizes the residual variance.

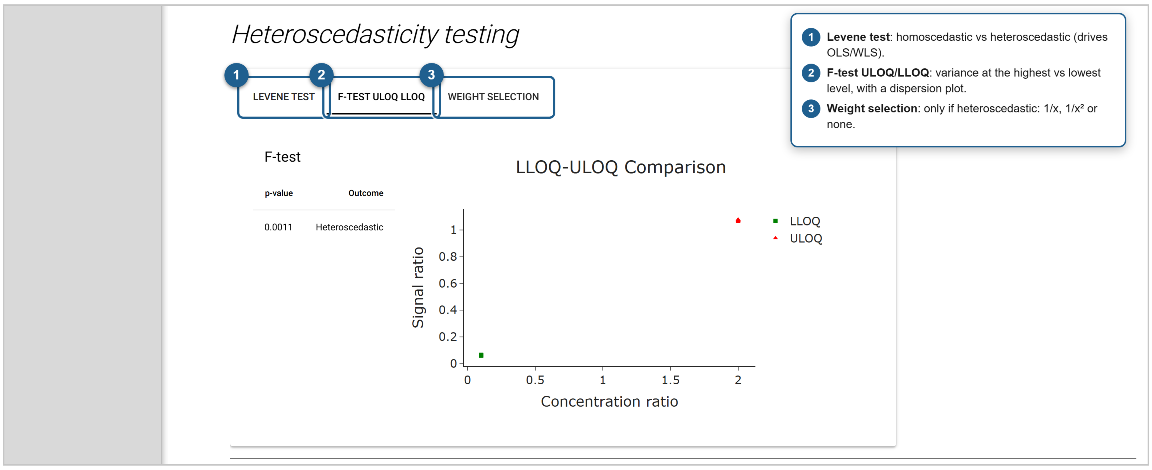

Figure 8. Heteroscedasticity testing tabs, showing the F-test result and the ULOQ-LLOQ dispersion plot.

Finally, the most suitable calibration model is presented in the last section of the page (Figure 9). Depending on the outcome of the tests, the model is fitted with Ordinary Least Squares (OLS) or Weighted Least Squares (WLS); the section header states which one is used. The **Best calibration model** dropdown 1 lets the user switch between the automatically selected model and the individual *Linear* and *Quadratic* fits. For each choice, the app reports the model parameters, Mandel test, and the calibration plot with the fitted equation and R^2.

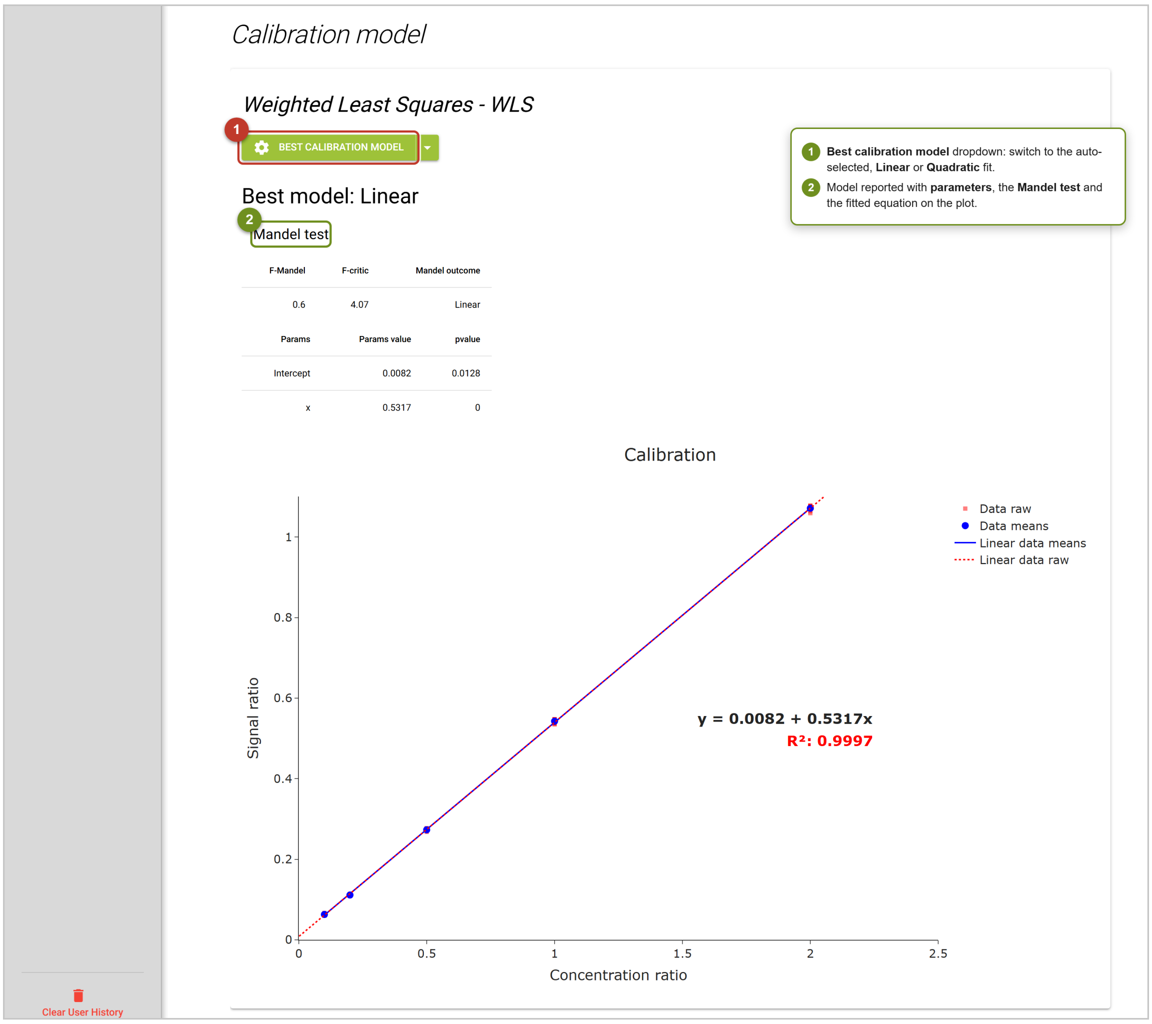

Figure 9. Automatically selected calibration model, with parameters, Mandel test and fitted equation.

Double check on residuals

When Mandel's test favours the quadratic model but a further t-test and F-test (at $\alpha = 0.01$) find the two models' residuals statistically indistinguishable, MVA overrides Mandel and keeps the simpler (linear) model. A dedicated card explains this whenever it happens.

Selecting *Linear model* or *Quadratic model* from the dropdown opens a detailed diagnostic view (Figure 10): the **Shapiro-Wilk test** on the residuals 1 with a pass/fail note, followed by the residuals-vs-concentration plot, the residual kernel density estimate (KDE) and the Q-Q plot 2.

4.3 Limits of Detection and Quantification

The decision limit (CCα), the limit of detection (LOD, reported as CCβ) and the limit of quantification (LOQ) are evaluated on a separate page (Figure 11) using the Hubaux and Vos method in the CCα/CCβ formulation of del Río Bocio et al. (J. Chemometrics 17, 2003). The algorithm requires a few parameters, which are set to generally optimal defaults but can be adjusted:

- the **statistical significance level** α 1, defaulting to 0.05;
- the **number of calibrators** 2 used by the algorithm (a radio with 3 or 4 levels, shown only when the dataset carries more than three calibration levels);
- the option to **use the first calibration point as LOQ** 3.

Two helper buttons 4 sit next to this last option: an **info** button that opens a short glossary of CCα, CCβ and LOQ, and a **book** button that displays Figure 2 of the source paper illustrating the geometry of the method.

CCα and CCβ

CCα (decision limit) is the lowest signal that can be told apart from the blank at risk α. **CCβ** is the LOD proper: the lowest concentration reliably detected, read off the lower calibration band. The LOQ is taken as twice CCβ unless the first calibration point is adopted instead.

Reducing the number of calibrators leads to a lower CCβ (LOD), and increasing the significance level (e.g. from 0.05 to 0.10) also lowers it. The computed limits are shown as a small table (CCα, CCβ and LOQ) next to two prediction-interval plots: the *Prediction interval* over the full calibration range and a zoomed *intercept region* view, in which the CCα and CCβ read-off markers are drawn on the lower prediction band (reproducing Figure 2 of the paper). By default the LOQ is estimated as twice CCβ; however, we recommend adopting the first calibration point as the LOQ, provided that its precision and accuracy meet the acceptance criteria.

Conservative by design

CCα and CCβ are computed from the *unweighted* (OLS) fit even when the calibration itself is weighted. This keeps the estimates conservative, since a weighted fit would shrink the residuals near zero concentration and report optimistically low limits.

If the computed CCβ (LOD) or LOQ exceeds the first calibration point, a warning is raised:

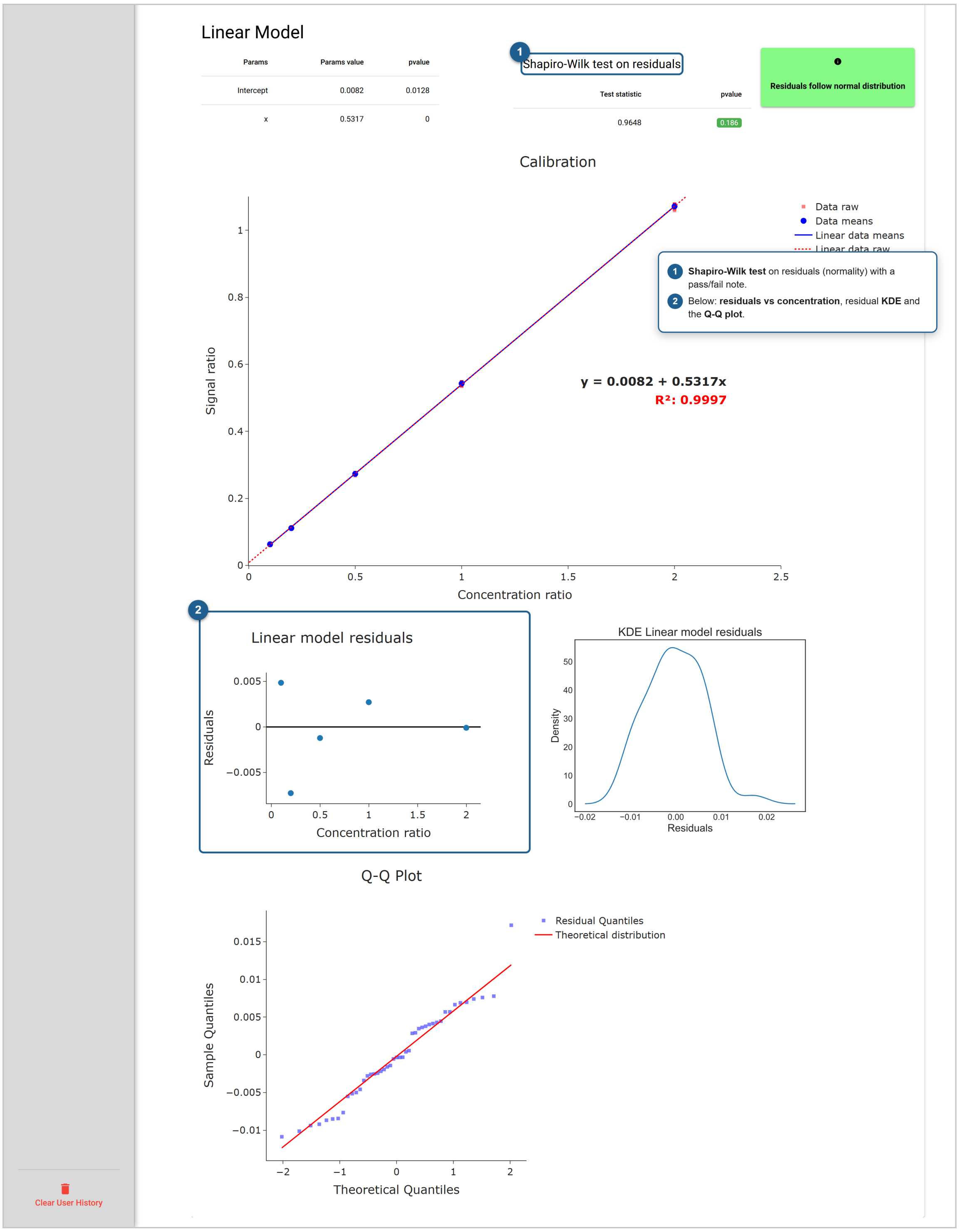

Figure 10. Linear-model diagnostics: Shapiro-Wilk test, residual plots, KDE and Q-Q plot.

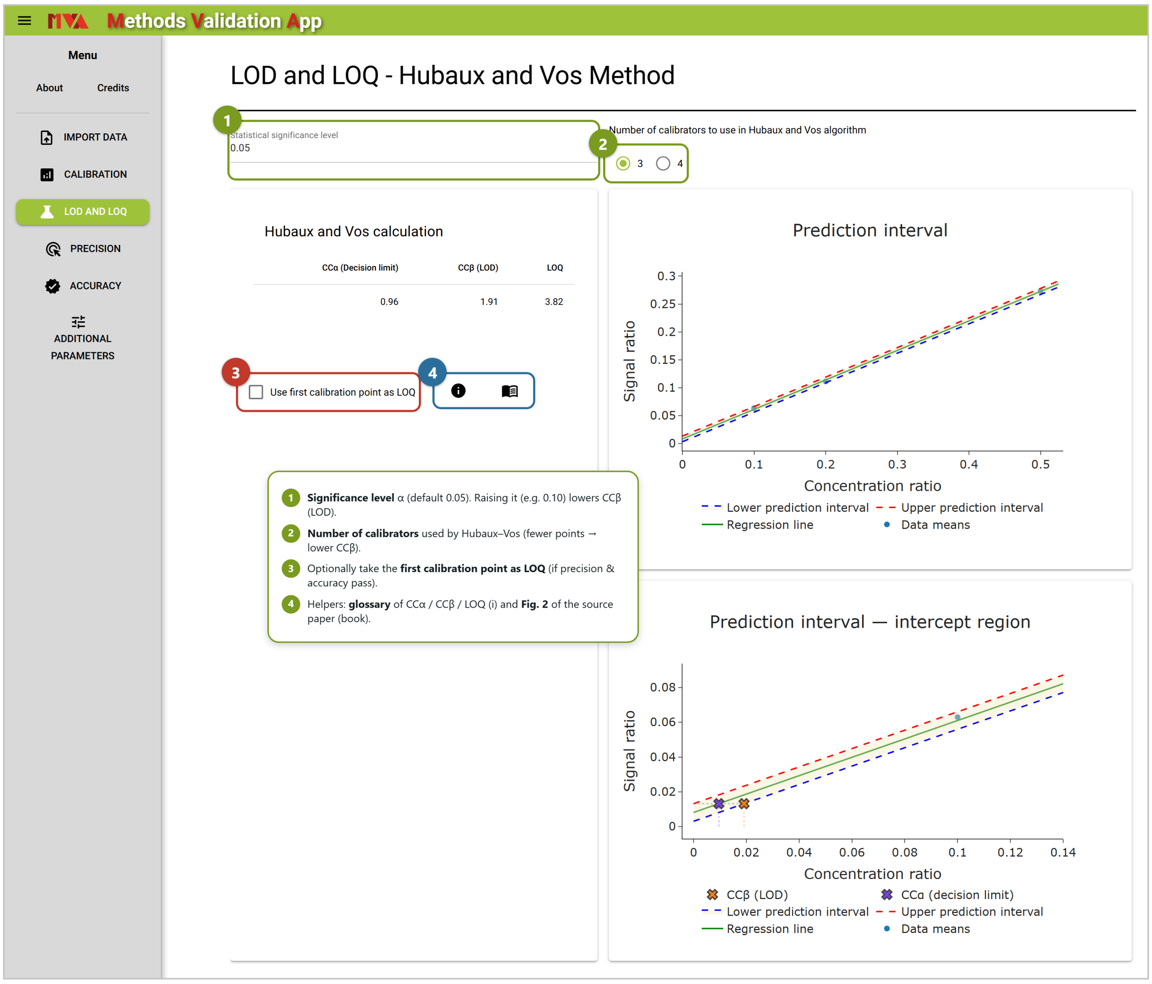

Figure 11. LOD and LOQ page: input parameters, the CCα/CCβ/LOQ table and the two prediction-interval plots (full range and intercept region).

this typically corresponds to a Type II error (false negative), meaning the working range does not extend low enough to support the estimated limit.

4.4 Precision and Accuracy

Precision and accuracy are evaluated on dedicated pages that share the same interface (Figure 12 shows the Precision page; Accuracy is identical). To start the evaluation the user enters the **number of calibration curves prepared per day** 1 (the same number must be used for all days) and the **number of validation days** 2. An expandable **acceptability criteria** panel 3 lets the user tune the colour thresholds with two sliders (default: optimal $\leq 20\%$ in green, acceptable $\leq 25\%$ in yellow, otherwise red).

Recommended design

For full compliance with the statistical framework, use **nine calibration curves over three days** (three replicates per day). A faster alternative is six curves over two days (three per day). If every curve is generated on a single day, only intra-day precision and accuracy are computed.

Both metrics are obtained by leave-one-out back-calculation. Intra-day precision/accuracy leaves out one curve at a time, builds the model on the remaining curves of the same day and predicts the left-out curve; inter-day leaves out one whole day at a time. The CV% (precision) or bias% (accuracy) is then computed level by level. Results are displayed in colour-coded tables that highlight the values meeting the selected criteria. A help button (top-left of each page) summarizes the calculation.

4.5 Additional Parameters

The *Additional parameters* page (Figure 13) covers two complementary quantities through separate tabs: **Matrix Effect** 1, which quantifies the ionisation suppression or enhancement caused by the sample matrix, and **Recovery** 2, which measures the extraction efficiency. Unlike the other modules, this page uses its own dedicated input files.

The workflow is identical for both tabs. The user chooses the number of **levels** and **replicates** 4, downloads the corresponding pre-formatted template, fills it with the measured signals, and uploads it back. A convenience switch 3 skips straight to the upload step when the template has already been prepared. Once the file is processed, MVA displays the per-level table and the averaged result, and offers a button to download the computed values as an Excel workbook.

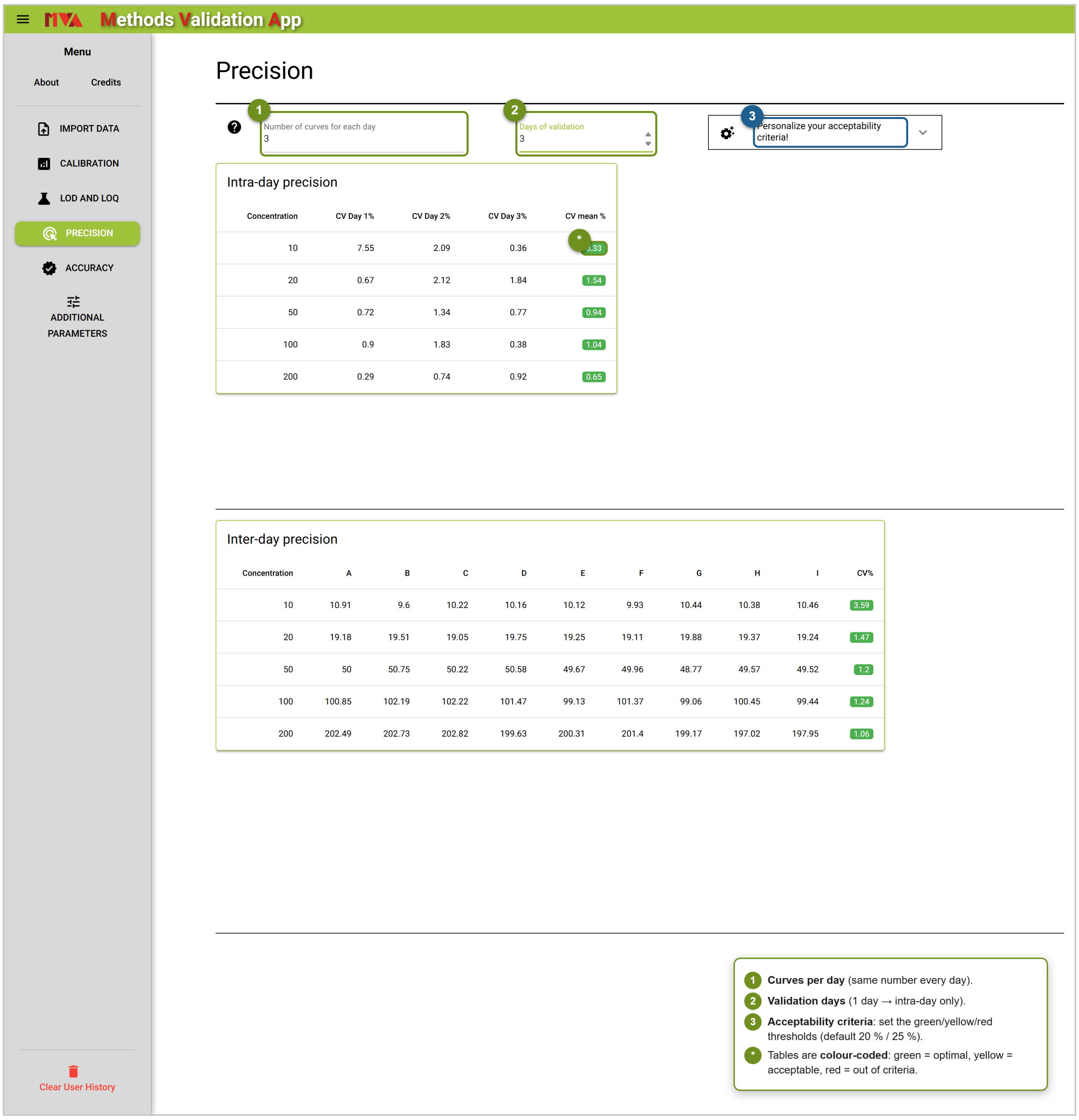

Figure 12. Precision page: design inputs, adjustable acceptability criteria and the colour-coded intra/inter-day tables.

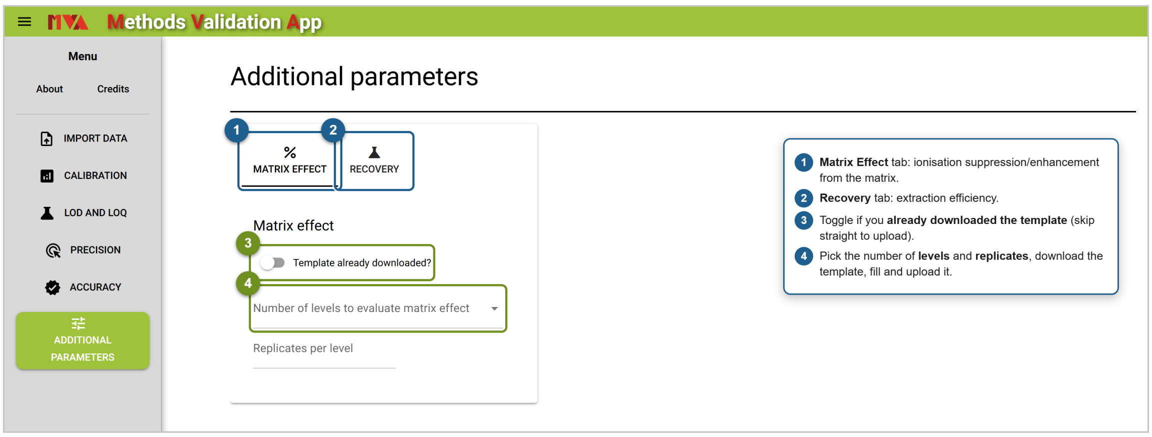

Figure 13. Additional parameters page: the Matrix Effect and Recovery tabs share a template-driven workflow.